

z1 (tn)

Rozłóż na czynniki liniowe wielomian $W(x) = (x^2 + 4x - 1)^2 - 16$ i podaj jego pierwiastki.

$$W(x) = (x^2 + 4x - 1)^2 - 16 \quad a^2 - b^2 = (a-b)(a+b)$$

$$W(x) = (x^2 + 4x - 1)^2 - 4^2$$

$$W(x) = [x^2 + 4x - 1 - 4] \cdot [x^2 + 4x - 1 + 4]$$

$$W(x) = (x^2 + 4x - 5)(x^2 + 4x + 3)$$

$$\Delta = b^2 - 4 \cdot a \cdot c$$

$$x_1 = \frac{-b + \sqrt{\Delta}}{2a}$$

$$x_2 = \frac{-b - \sqrt{\Delta}}{2a}$$

$$x^2 + 4x - 5 = 0$$

$$x^2 + 4x + 3 = 0$$

$$\Delta_1 = 16 + 20 = 36$$

$$\sqrt{\Delta_1} = 6$$

$$x_1 = \frac{-4 + 6}{2} = 1$$

$$x_2 = \frac{-4 - 6}{2} = -5$$

$$\Delta_2 = 16 - 12 = 4$$

$$\sqrt{\Delta_2} = 2$$

$$x_3 = \frac{-4 + 2}{2} = -1$$

$$x_4 = \frac{-4 - 2}{2} = -3$$

Wyznaczamy pierwiastki tego wielomianu:

$$(x+5)(x-1)(x+3)(x+1) = 0$$

$$x+5=0$$

$$\underline{x = -5}$$

$$x-1=0$$

$$\underline{x = 1}$$

$$x+3=0$$

$$\underline{x = -3}$$

$$x+1=0$$

$$\underline{x = -1}$$

Odp: $W(x) = (x+5)(x-1)(x+3)(x+1)$,
pierwiastki to: $x \in \{-5, -3, -1, 1\}$.

Zatem :

$$W(x) = (x+5)(x-1)(x+3)(x+1)$$